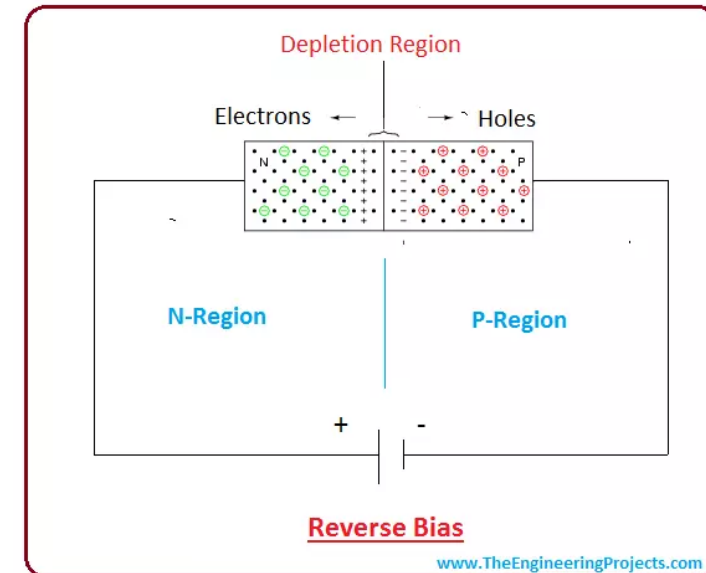
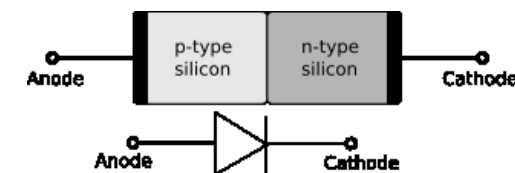


P-N Junction: A Diode

- Reverse bias creates the opposite effect in comparison to forward bias, forcing the majority carriers of each doped region to move away from the contact region, and thus increases the depletion region's size.
- This behavior leads to the conclusion that the p-n junction forms a diode.
- The small current flowing under the reverse bias is known as leakage current.
- Of course, this holds for voltage that is not excessive. After a particular voltage value, the p-n junction will essentially collapse and start conducting even if it is reverse-biased. The abundance of electrons (charge carriers) in the p-doped region will cause it to become more n-charged than the n-doped region, causing a breakdown of the p-n junction's ability to prevent current from running through it when it is reverse biased.



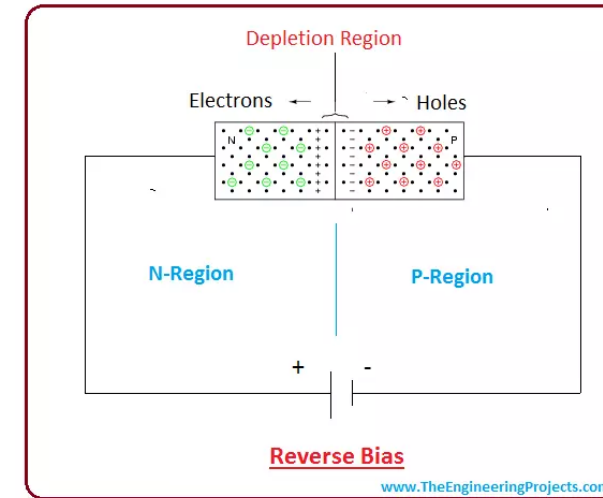
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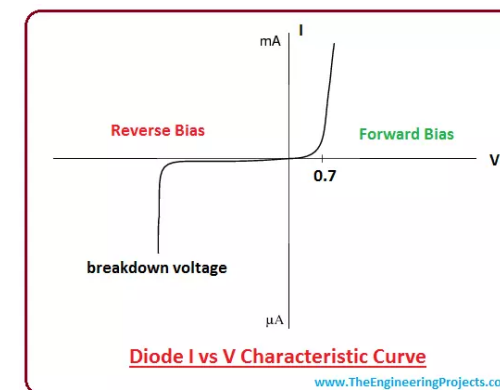
https://upload.wikimedia.org/wikipedia/commons/thumb/7/79/PN_diode_with_electrical_symbol.svg/320px-PN_diode_with_electrical_symbol.svg.png

P-N Junction: (Reverse) Breakdown

- When the voltage exceeds the breakdown value, the depletion region will quickly reduce in size and the current will suddenly increase.
- Essentially a short-circuit will have taken place.
- A similar situation occurs in forward-biasing, and the p-n junction's mechanism completely collapses, when the voltage provided exceeds the built-in potential (which is around 0.6 to 0.9 V for Si).

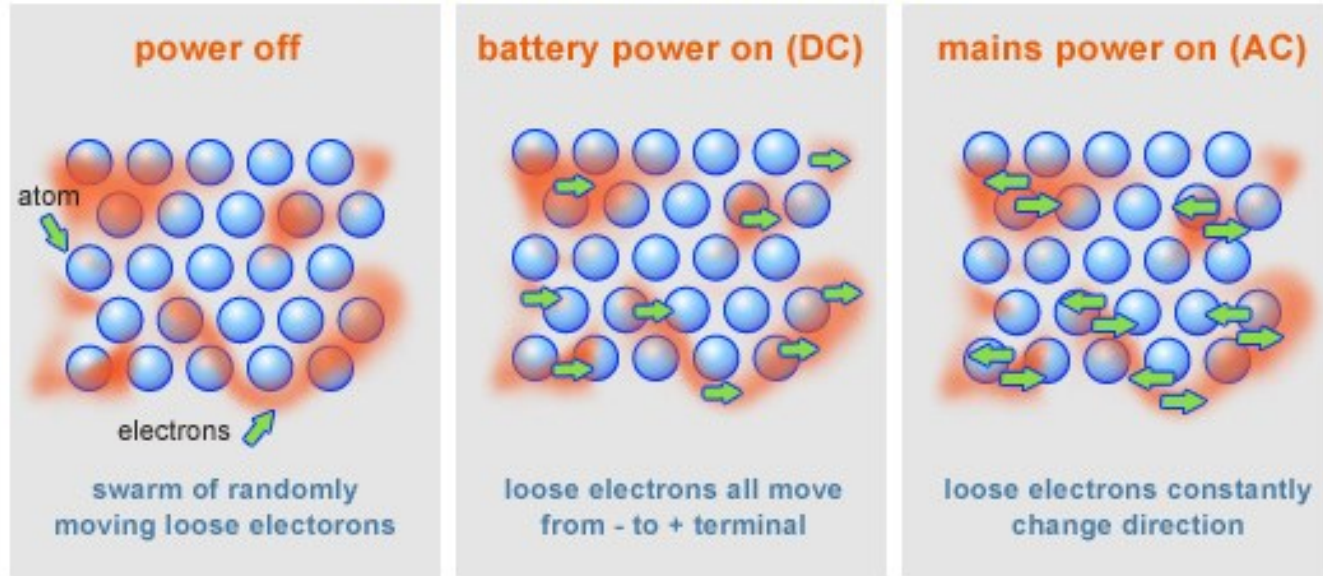


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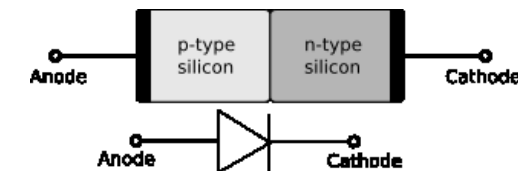
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What goes on in electrical wires

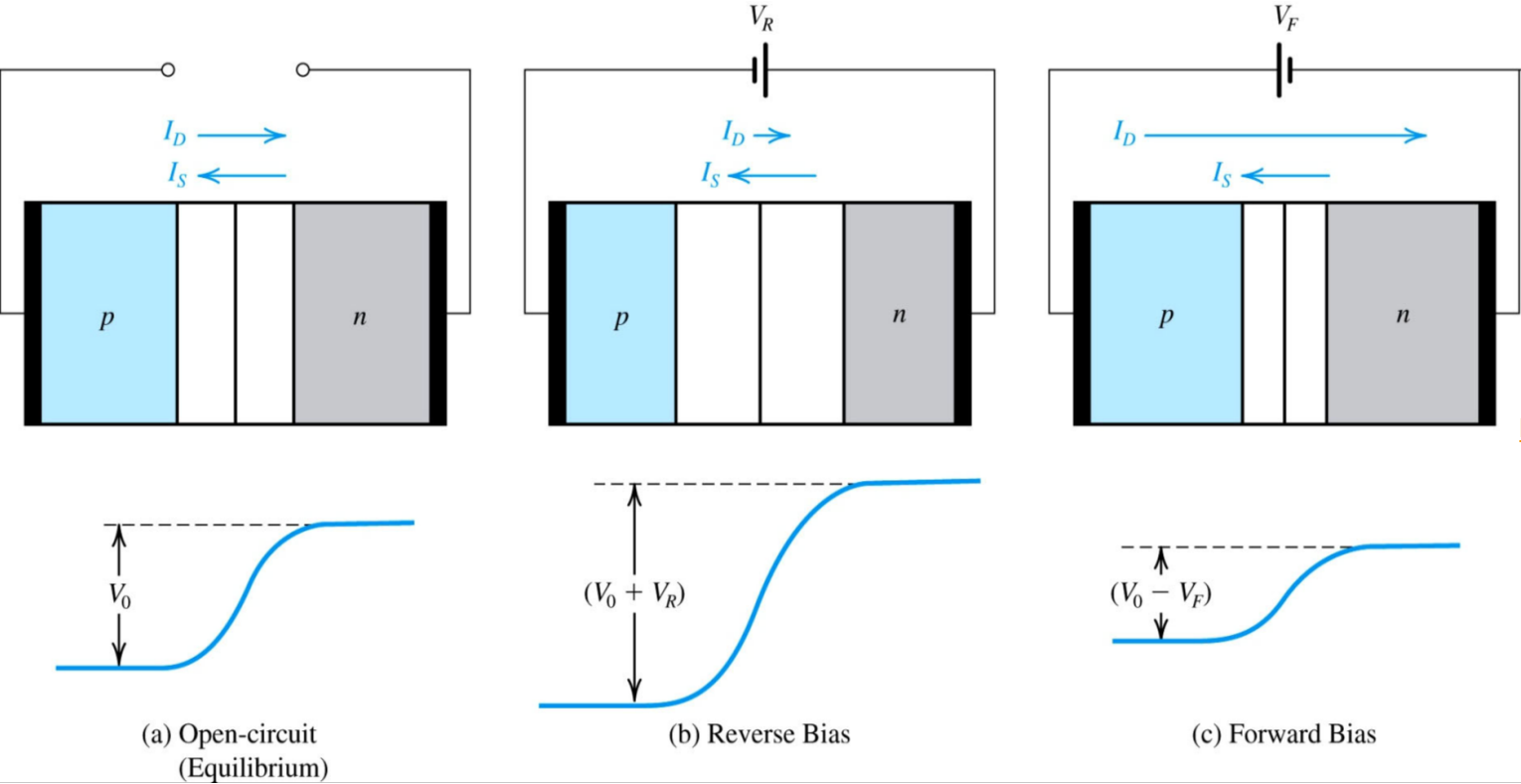


If you could see what's going on at the atomic level inside a wire, it'd look something like this. Like all metals, wire is made of rigid framework of atoms with a swarm of loose electrons buzzing around them. Hook a wire up to a battery (DC) and the electrons all move forwards in synch. Plugged into a power point the electrons in a wire jiggle forwards and backwards on the spot, changing direction 100 times per second (AC).

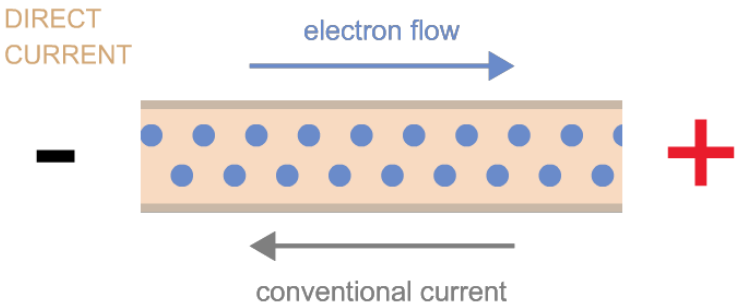
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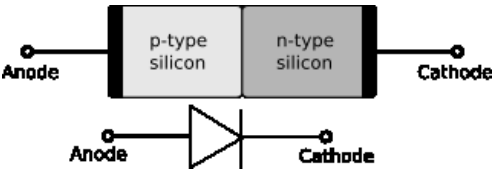
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Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.



<https://serpmedia.org/scigen/images/electrons-dc%20copy.gif?crc=57018027>



https://upload.wikimedia.org/wikipedia/commons/thumb/7/79/PN_diode_with_electrical_symbol.svg/320px-PN_diode_with_electrical_symbol.svg.png

P-N Junction: Water Canal with Dam of Rocks

- **Analogy to water**

- P-N junction: Canal with a dam of rocks (depletion region)
- Dam of rocks allows some water leakage (leakage current)
- With forward bias, towards the existing flow, some rocks are removed (the depletion region becomes smaller)
- With reverse bias, against the flow, more rocks are added to the dam (the depletion region becomes bigger)
- With too much reverse bias, the rocks that are added carry away the existing rocks, the dam breaks and there is a (reverse) flood in the canal
- With too much forward bias, there is also a flood in the regular direction of flow.

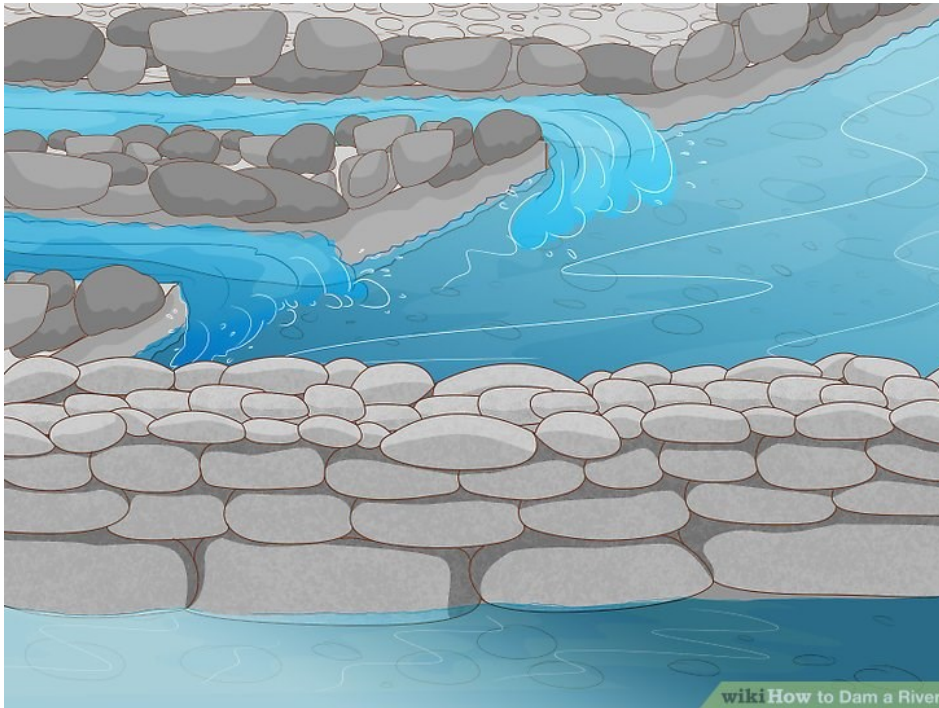


<https://assets-news.housing.com/news/wp-content/uploads/2022/10/14070049/CANAL-IRRIGATION-FEATURE-compressed.jpg>



<https://assets.thehansindia.com/h-upload/2021/09/04/1106050-river-waters.webp>

P-N Junction: Water Canal with Dam of Rocks



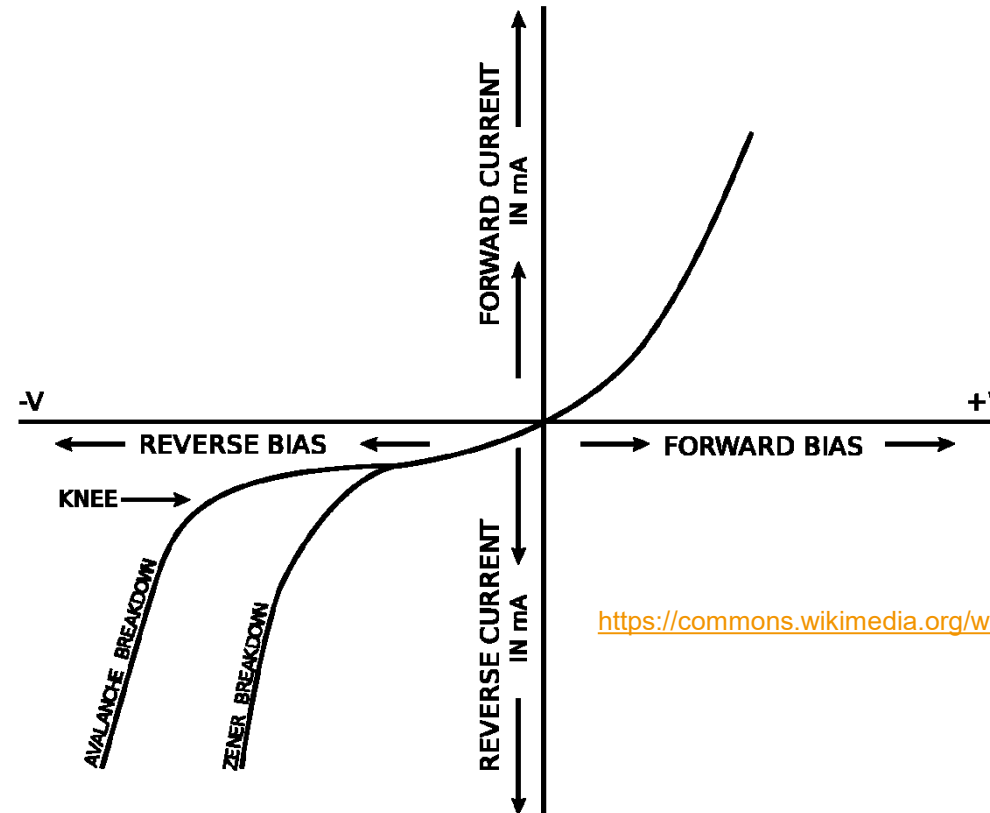
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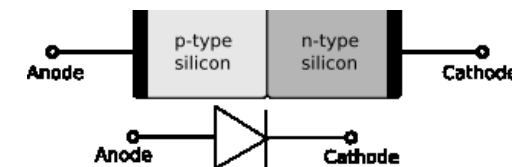
<https://www.wikihow.com/Dam-a-River#/Image:Dam-a-River-Step-5-Version-2.jpg>

Zener and Avalanche Breakdowns

- Zener breakdown occurs in a Zener diode at low reverse voltages.
- A Zener diode is heavily doped to reduce the breakdown voltage.
- This causes a very thin depletion region.
- As a result, an intense electric field exists within the depletion region.
- Close to the Zener breakdown voltage, the field is sufficiently able to pull electrons from their valence groups and create current.



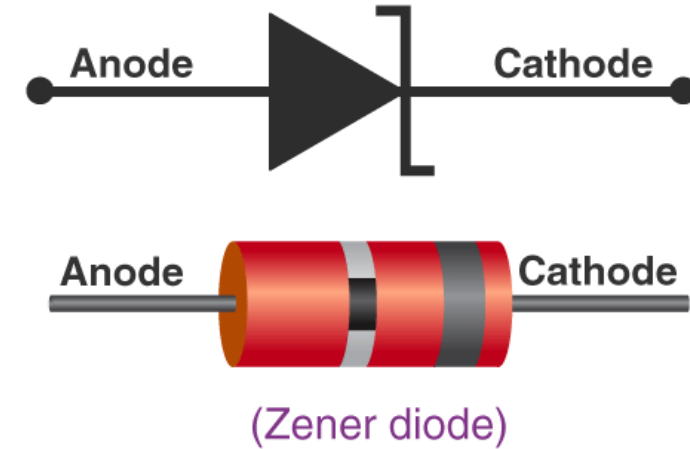
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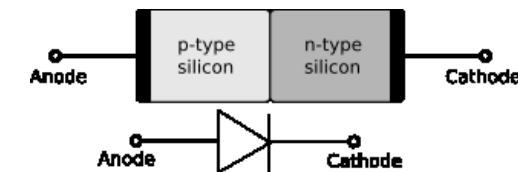
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Zener and Avalanche Breakdowns

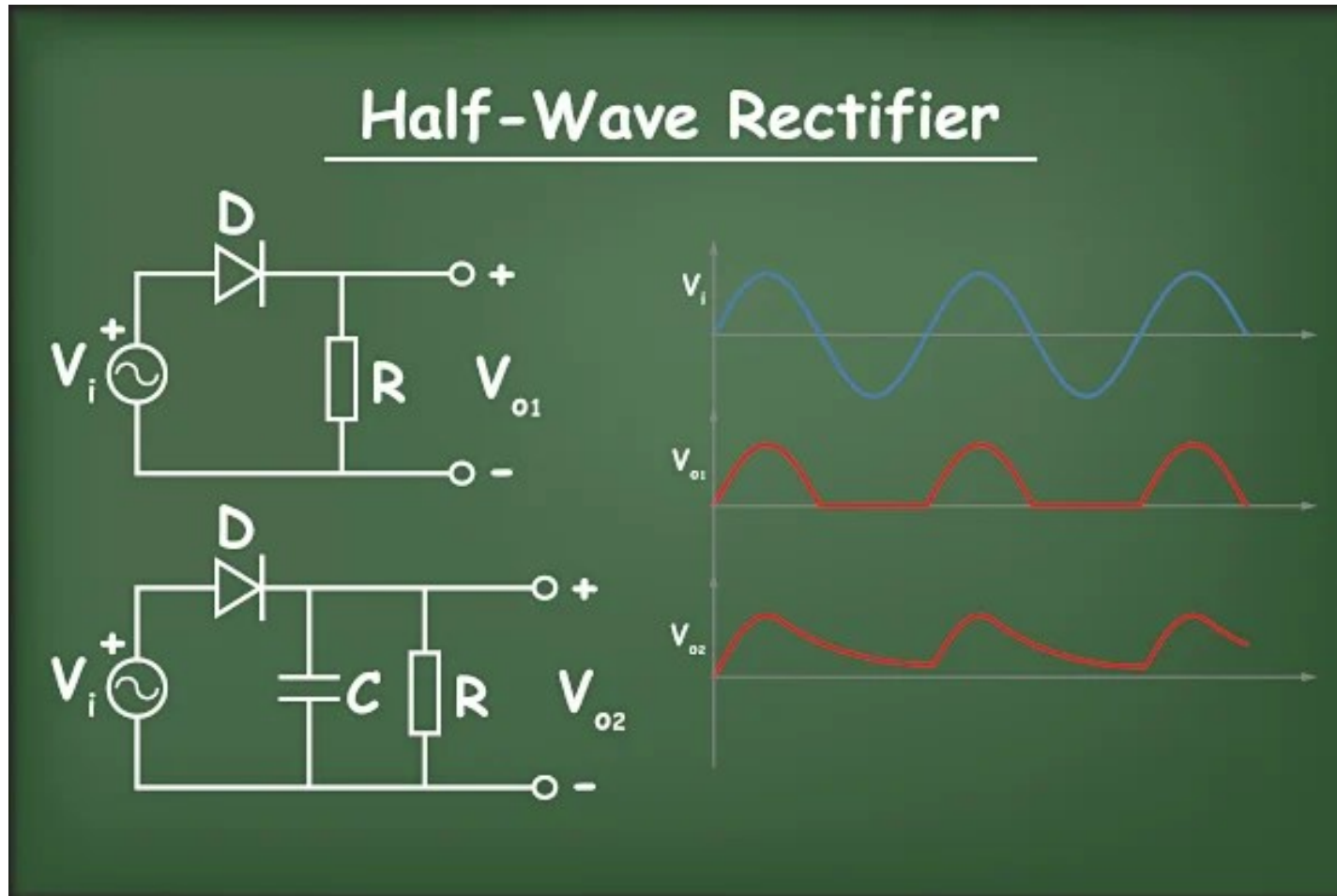
- The Zener breakdown occurs due to high doping that leads to a narrow depletion region.
- It is roughly based on quantum tunnelling
- “It occurs in a reverse biased p-n diode when the electric field enables tunnelling of electrons from the valence to the conduction band of a semiconductor, leading to numerous free minority carriers which suddenly increase the reverse current.”
(https://en.wikipedia.org/wiki/Zener_effect)



<https://cdn1.byjus.com/wp-content/uploads/2022/12/Zener-diode-2.png>

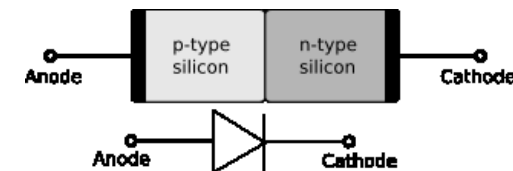


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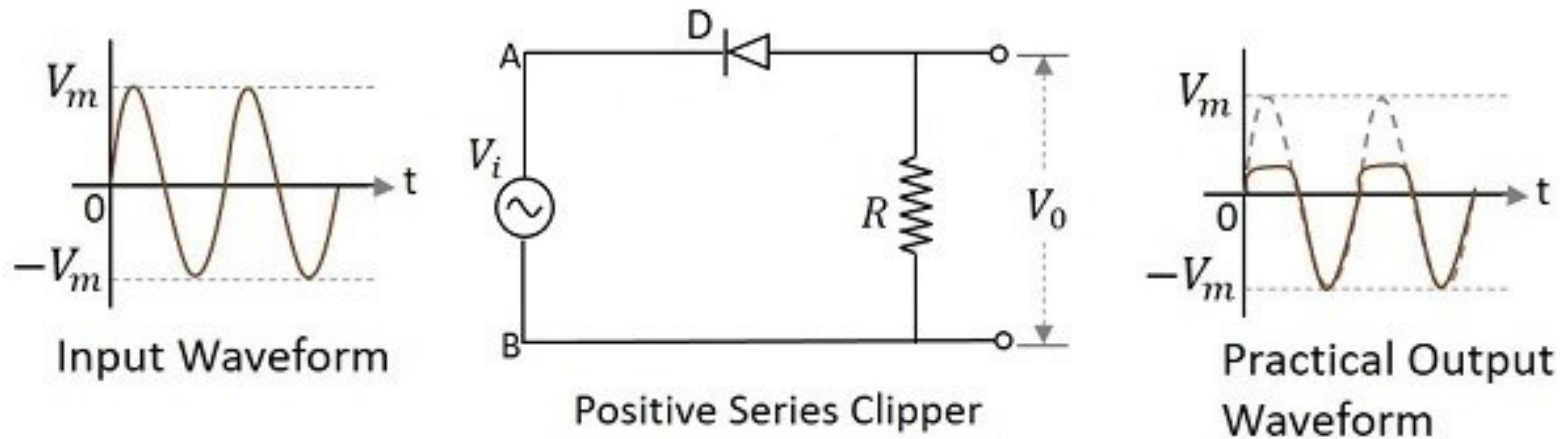


Clipper / Clipping circuit

https://stock.adobe.com/ph/images/half-wave-rectifier-circuits-with-capacitor-and-diagram-on-chalkboard-vector/100717226?asset_id=100717226

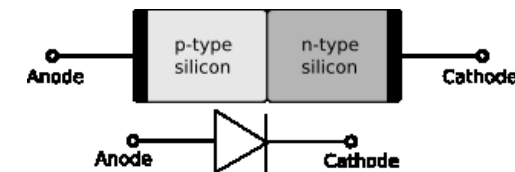


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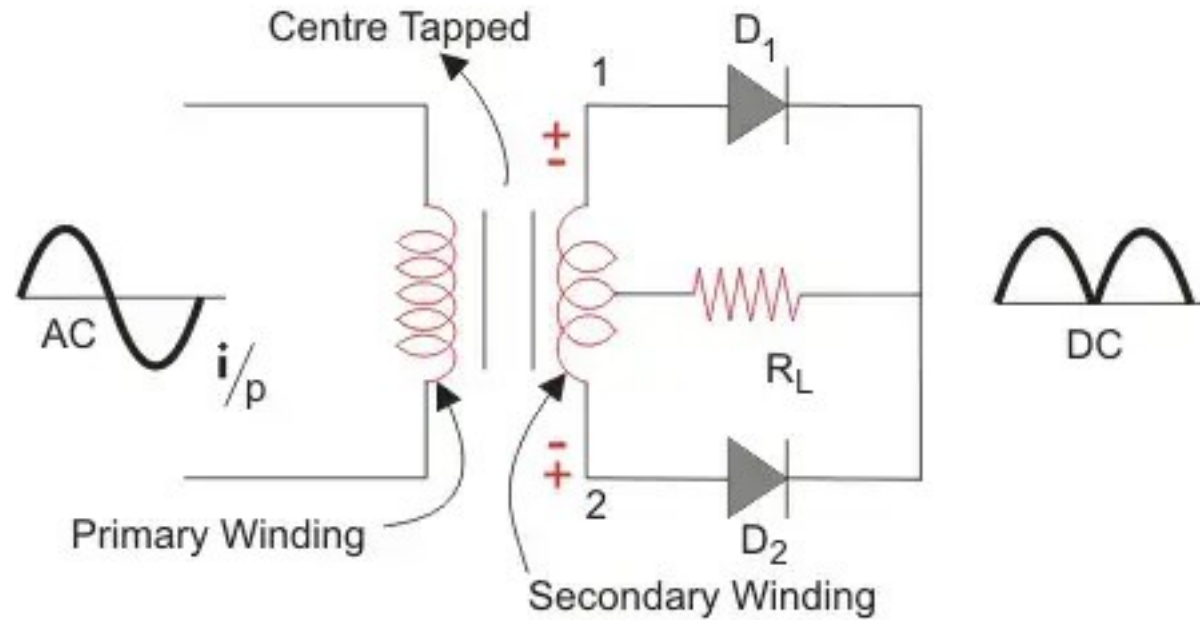
Clipper / Clipping circuit

https://www.tutorialspoint.com/electronic_circuits/electronic_positive_clipper_circuits.htm



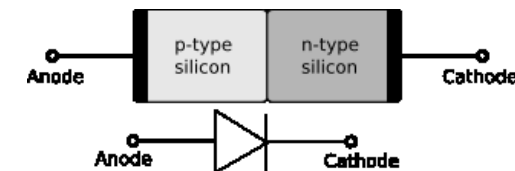
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Applications of Diodes: Rectifiers



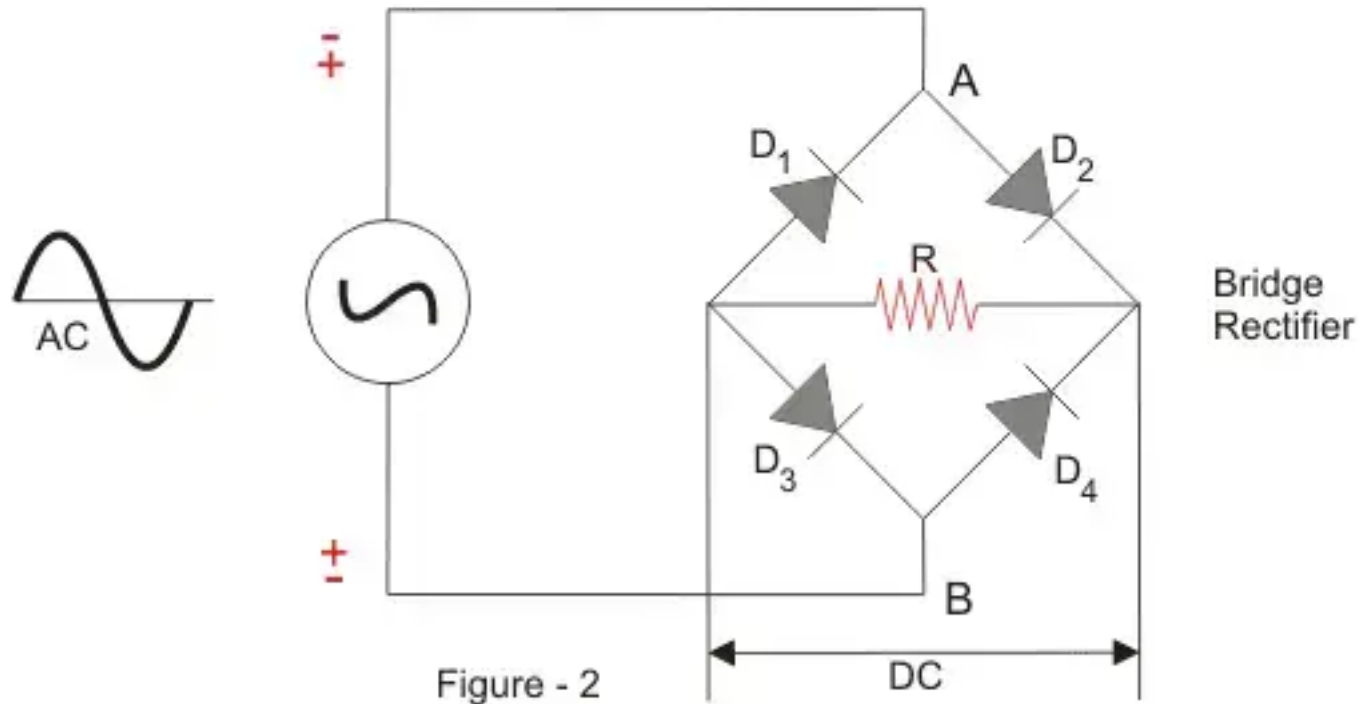
Centre Tapped Full Wave Rectifier
Figure - 1

<https://www.electrical4u.com/images/2018/march18/1521538453.png?ezimgfmt=rs:498x302/rscb38/ng:webp/ngcb38>

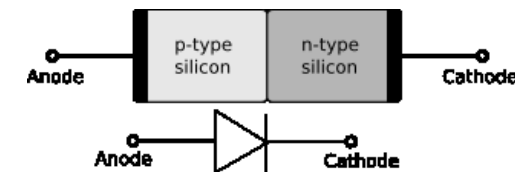


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Applications of Diodes: Rectifiers



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